

FME Prototype Extraction and Migration Plan for OpenEd

Purpose

This document explains how to turn the current Frontier Model Evaluations course prototype into OpenEd v1 without throwing away the useful work.

What to keep

The current FME prototype contains reusable assets:

- Course shell.
- Course route structure.
- Lesson runtime concepts.
- BYOK coach panel.
- Video reference player.
- Source library.
- Glossary.
- Evidence card library.
- Artifact builders.
- Capstone studio.
- Content QA screen.
- Design token work.
- Course content data.

What to change

Current issue	OpenEd migration fix
FME-specific labels	Generalize to course/runtime vocabulary.
Dark theme too heavy	Light-first OpenEd theme and calmer dark mode.
Lesson screens repetitive	Lesson type variants.
Flat UI	Add visual learning moments and progress maps.

Current issue	OpenEd migration fix
Capstone weak but usable	Keep for prototype; rebuild later as generic project studio.
Hardcoded data	Move to course schema.
Course-only app	Add auth, learner, educator, proof surfaces.
BYOK coach prototype	Formalize tutor modes and storage/security.

Refactor plan

Step 1: Rename and reframe

- App name: OpenEd.
- FME becomes first course.
- Routes under `/course/:courseSlug/...`
- Components under generic names.

Step 2: Extract course runtime

Create:

- CourseShell.
- CourseDashboard.
- LearningPath.
- LessonWorkspace.
- SourceDrawer.
- TutorPanel.
- QuizRuntime.
- ArtifactRuntime.
- PortfolioRuntime.

Step 3: Add platform shell

Create:

- Public landing.
- Auth pages.
- Learner dashboard.
- Course catalog.
- Educator Studio.

- OpenEd Team console.

Step 4: Build data model

Move from local course data to Supabase tables:

- courses.
- modules.
- lessons.
- sources.
- enrollments.
- progress.
- artifacts.
- rubrics.
- tutor_sessions.

Step 5: Fix design system

- Light mode default.
- Dark mode softened.
- Phone/tablet/desktop breakpoints.
- New OpenEd brand tokens.
- Course-specific accent colors.
- More visual diagrams and milestones.

Step 6: Capstone treatment

For v1:

- Keep FME capstone as prototype project.
- Label as "FME Capstone Prototype."
- Do not present as the universal OpenEd capstone model.

Later:

- Build generic Project Studio template.
- Add reviewer defense.
- Add stronger quality gates.

Component mapping

FME component	OpenEd component
BYOKCoachPanel	AITutorPanel
VideoReferencePlayer	LessonMediaPlayer
EvidenceCardBuilder	ArtifactBuilder / ProofCardBuilder
BenchmarkPacketBuilder	Course-specific ArtifactTemplate
RiskDashboardBuilder	ProjectDashboardBuilder
CapstoneStudio	ProjectStudio
ContentQAScreen	PublishReadinessQA
SourcesScreen	SourceLibrary
GlossaryScreen	Glossary

Migration risks

Risk	Mitigation
Break existing FME flow	Keep FME as seeded course data and test routes.
Overgeneralize too early	Generalize only shared runtime primitives.
Add Educator Studio before learner runtime stabilizes	Build in parallel but release learner first.
UI remains flat	Create lesson variants and visual excitement kit.
Mobile is an afterthought	Build phone lesson and tutor flow early.

Completion criteria

Migration is successful when:

- FME works as a course inside OpenEd.
- Learner app is not FME-branded only.
- Educator can draft a new course using the same schema.
- Artifacts flow into SkillProof portfolio.
- OpenEd Team can review course QA.
- Responsive UI is acceptable on phone, tablet, desktop.

Source notes

This document is grounded in the current OpenEd/FME project context and the following reference set where applicable:

- Uploaded Frontier Model Evaluations prototype code bundle, latest inspected bundle: Design System for Frontier Model Evaluation (5).zip.
- Uploaded Frontier Model Evaluations 51-hour curriculum and Module 1 foundations files.
- Coursera, Udemy, DataCamp, Moodle, Khanmigo, Duolingo public documentation and product pages.
- Supabase Auth and Row Level Security documentation.
- OpenAI API key safety guidance, MDN Web APIs, and YouTube IFrame Player API documentation.
- Bloom's 2 sigma problem, intelligent tutoring systems research, CourseAssist, LearnLM-Tutor, DCCI, and UNESCO guidance for generative AI in education.

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